

AIS Graph Analytics

From vessel positions to port networks and AI-assisted anomaly insight.

Current repository architecture and exported three-tab dashboard

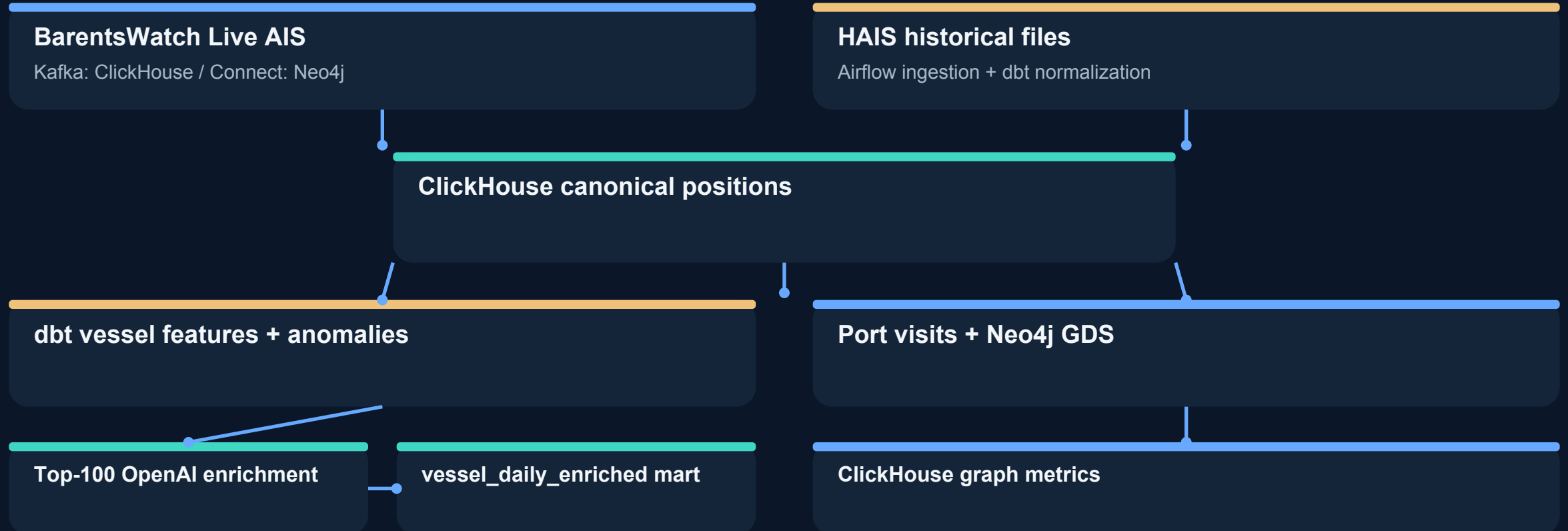
ONE DAILY MASTER

TOP 100 AI ELIGIBLE

THREE DASHBOARD TABS

Two ingestion paths, one analytical platform

Airflow coordinates daily analytics; ClickHouse stores the canonical AIS history.



Metabase: Ports | Vessels | Anomalies & AI Insights

Airbyte is optional for the core pipeline. Country-enriched port views need a separate country sync.

The combined daily_ais_pipeline

Scheduled at 02:00 UTC. Defaults to yesterday UTC; accepts a historical activity_date.

01 Resolve activity_date

02 dbt core + daily features

03 Port visits

04 Neo4j GDS

05 Graph metrics export

06 dbt graph models

07 Daily anomalies

08 AI enrichment

09 Final vessel mart + tests

One captured UTC date pins the port-visit window and AI selection. dbt builds historical relations.

The three graph DAGs remain independently runnable for manual diagnosis. The master waits for each successful child.

Deterministic detection, bounded AI explanation

dbt decides which vessel-days are anomalous before any OpenAI call.

Daily features

Per vessel and activity_date: coverage, speed, stationary share, and navigation status.

Seven-day baseline

Eligible prior days for the same vessel. At least 12 observation hours and 100 AIS points; three usable baseline days per metric.

Anomaly score + rank

Speed and stationary deviations drive the relative severity score. Status signals add small increments. Rank resets each day.

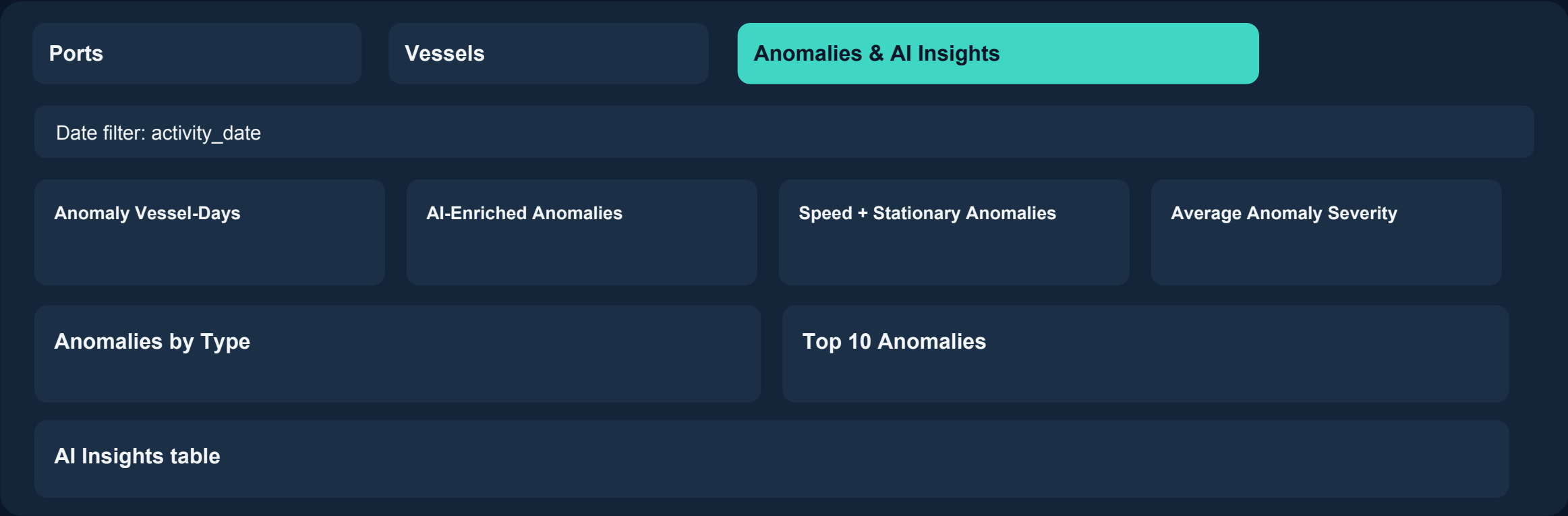
AI history and final mart

Only anomaly_rank ≤ 100 is eligible. Matching model, vessel_anomaly_v1 prompt, and input hash are skipped. New results append to vessel_ai_enrichment; dbt joins them into vessel_daily_enriched.

AI_ENRICHMENT_LIMIT caps new calls after already-enriched candidates are skipped. The final mart may include rows without AI content.

One dashboard, three analytical views

Schematic based on the tracked Metabase JSON export. This is not a live screenshot.



AI KPI: has_ai_enrichment = 1, ai_prompt_version = vessel_anomaly_v1, anomaly_rank <= 100. Review fixed historical Date and Run ID defaults after import.

What each layer owns

The core daily path and optional enrichments have distinct setup requirements.

ClickHouse

Canonical raw AIS, AI response history, and dated graph snapshots.

dbt

Vessel features, deterministic anomalies, final AI-enriched mart, and port views.

Neo4j GDS

Observed port links, weighted PageRank, and Louvain communities.

Airflow

Shared date, child DAG sequencing, enrichment, and tests.

Metabase

Three-tab presentation from built ClickHouse models.

Airbyte (optional)

REST Countries sync for country-enriched port views; not required by the core master.

Verified history and current scope

Past acceptance results stay labeled as historical evidence.

24 SEP 2026 / HISTORICAL ACCEPTANCE

24 Sep 2026 acceptance: 2,941,803 source positions, 2,696 visits, and 51 country-enriched port metric rows for the 23 Sep UTC window. These figures are not current dashboard counts or a throughput benchmark.

LIMITS / NEXT VALIDATION

Known limits: daily visit detection can split a stay across midnight; bounded in-memory batches limit scale. Backup/restore and isolated fresh-install validation remain deferred.

Repository references: README.md; airflow/README.md; dbt/README.md; metabase/README.md; airflow/dags/daily_ais_pipeline.py; dbt/models/marts/vessel_daily_anomalies.sql; metabase/exports/norway-port-graph-analytics/dashboard.json.